

Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs <- rhfs |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSpace, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECOMMREN, RNTAPPFEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
  )

# Renaming some factor levels for later convenience
rhfs <- rhfs |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
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    "Assisted living limited interview; self response" = "3",
    "Conversion from partial to full interview" = "5",
    "Interview Compl; field interview" = "201",
    "Assisted living limited interview; field interview" = "203",
    "Sufficient partial; field interview" = "204",
    "Military limited field interview" = "206"
  )
)

# Rearranging variables greatly improves View(rhfs)
questionnaire_order <- c(
  # 1c-1e and property definition
  "PROPTOWN", "PROPCON", "COMPLEX",
  "NUMBLD_R", "NUMUNITS_R", "NUMCAT_R",

  # 2-4e
  "UNITSTAT", "YRNEWBLG_R", "HOWACQ", "YRPROPACQ_R", "PURPRICE_R",
  "CASHDWN", "CASHDWNAMT", "MRKTVAL_R", "HOWMVAL", "APPRWAIVER",

  # 5-6: occupancy, rents, and rent payment methods
  "BROOMS", "TOTRENTVAC", "TRENOC_R", "TRENVA_R", "TOWNOC_R",
  "TEMPOC_R", "TOTHVA_R", "LOWRENT", "HIRENT", "RENT",
  "RENTNO_R", "RENTLT249_R", "RENT274_R", "RENT299_R", "RENT349_R",
  "RENT399_R", "RENT449_R", "RENT499_R", "RENT549_R", "RENT599_R",
  "RENT649_R", "RENT699_R", "RENT749_R", "RENT799_R", "RENT899_R",
  "RENT999_R", "RENT1099_R", "RENT1249_R", "RENT1499_R", "RENT1999_R",
  "RENT2499_R", "RENT2999_R", "RENT3499_R", "RENT3999_R", "RENT4499_R",
  "RENT4999_R", "RENT5000_R", "BED0_R", "BED1_R", "BED2_R", "BED3_R",
  "PAYONLINE", "PAYTHIRD", "PAYAPP", "PAYCREDIT", "PAYCHK", "PAYCASH", "PAYOTH",

  # 6a-11c
  "LEASLNG", "LNGTRMPRV", "WEEKLNG", "WEEKLNGNUM", "WEEKLNGPCT",
  "RNTAPPFEE_R", "NORNTAPPFEE", "RNTAPPFEewaiv", "PCTFEewaiv_R", "SECDEPAMT",
  "CONCSS", "CONCSSNUM", "SENIOR", "RCONTROL", "RCONTROLAHS",
  "OFFSTPARK", "OFFSTPARKAHS", "IELEC", "IGAS", "IWATER", "ISEWER", "ITRASH",
  "ICABLE", "IWIFI", "IPOOL", "IPARKING", "PROPANS",

  # 12a-15e
  "TOTCOLL", "COMMSpace", "RECOMMREN", "COMMRENT", "MNGMNT", "HRSMNGMNT",
  "ADVERTISE", "ADWEBSITE", "ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
  "ADAGENCY", "ADAGENT", "MLS",

  # 16: operating expenses
  "OPPAY", "OPTAX_R", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
  "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX", "OPEX_R",

  # 17-18c: capital expenses and ownership
  "CAPEX_R", "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT", "CPACCAMT",
  "CPOTHAMT", "CPRESAMT_R", "OWNENT", "OWNOTHPROP", "NUMOTHPROP", "NUMOTHUNITS",

  # 19-23: subsidies, tax credits, and tax reductions
  "HUDUNITS", "GOVBELow", "RSUBSIDY", "ELDERLY", "GOVGRANT", "BENEOTH",

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"LIHTCUNITS", "STLOWINC", "HISTORIC", "LOWINCTEN", "LOWINAREA", "ECONDEVLP", "OTHRVCV",

# 24-37: debt/mortgage section and HUD voucher history
"DEBT", "WHYNOMRT", "DEBTRNUM", "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R",
"APPVAL_R", "GREENMORT", "MORTGET", "LOCTYP", "REGPAY", "PMTAMT_R", "PMTFREQ",
"RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX", "ACCEPT"
)

rhfs <- rhfs |>
  relocate(
    CONTROLPUF, OUTCOME, SAMPSTRATA, SURVYR,
    all_of(questionnaire_order)
  ) |>
  # Keep the full-sample and replicate weights together at the far right.
  relocate(
    WEIGHT, starts_with("REPWGT"), .after = last_col()
  )

# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )

## # A tibble: 1 x 3
##   properties properties_se   moe
##   <dbl>         <dbl> <dbl>
## 1    18965.         165.  271.

# Makes View(rhfs) much more productive.
rhfs <- rhfs |>
  select(-starts_with("REPWGT"), -WEIGHT,
    -SURVYR, -CONTROLPUF, # is always 2024, ID isn't useful in data analysis
    # J flag variables mark values that were manually reviewed; note that

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    # they may become relevant if outliers are under scrutiny.
    -starts_with("J"),
    # A continuous variable coded as a factor without documentation; may
    # be useful for pure prediction
    # -PCTFEEWAIV_R,
    # Poorly documented variable with 99.5% of its values missing
    -WEEKLNGPCT
  )

# NAs in the dataset
missingvalues <- function(df) {
  NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
  NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

  tibble(
    NAs = NAs, # Not applicable
    NRs = NRs, # Not reported
    perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
  )
}
missingvalues(rhfs)

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## # A tibble: 1 x 3
##       NAs    NRs perc.missing
##   <int> <int>      <dbl>
## 1 170275 144234      0.402

```

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# Missing values by interview type
rhfs |>
  summarize(
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )

```

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##              OUTCOME    NAs    NRs perc.missing
## 1      Interview Compl; field interview 97135 73188 0.3933920
## 2      Sufficient partial; field interview 29948 36570 0.5177304
## 3      Interview Compl; self response 37769 21028 0.3230888
## 4      Conversion from partial to full interview 3442 7740 0.4538149
## 5      Military limited field interview 297 839 0.7171717
## 6 Assisted living limited interview; field interview 1286 3761 0.7169034
## 7 Assisted living limited interview; self response 398 1108 0.7130682

```

```

# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to them.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
    # In a given observation, if any of these columns contain a value other than

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    # -8 or -9 return True.
    if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
  ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1

# Same idea as with REGPAY
trivially.missing.ownothprop <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OWNOTHPROP == -9 &
    if_any(c(NUMOTHPROP, NUMOTHUNITS), \(x) !x %in% c(-8, -9))
  ) |>
  select(id, NUMOTHPROP, NUMOTHUNITS) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.ownothprop] <- 1

# Same idea as with REGPAY
trivially.missing.rntappfeewaiv <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    RNTAPPFEewaiv == -9 &
    if_any(c(PCTFEewaiv_R), \(x) !x %in% c(-8, -9))
  ) |>
  select(id, PCTFEewaiv_R) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.rntappfeewaiv] <- 1

# Ensuring that when CASHDWN is 2, CASHDWNAMT is always -9
rhfs |>
  mutate(id = row_number()) |>
  select(id, CASHDWN, CASHDWNAMT) |>
  filter(CASHDWN == 2 & CASHDWNAMT != -9)

```

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## [1] id          CASHDWN    CASHDWNAMT
## <0 rows> (or 0-length row.names)

```

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# If a CASHDWN payment wasn't made, then the CASHDWNAMT should be 0
rhfs$CASHDWNAMT <- ifelse(rhfs$CASHDWN == 2, 0, rhfs$CASHDWNAMT)

# Same idea as with CASHDWN/CASHDWNAMT
rhfs |>
  mutate(id = row_number()) |>
  select(id, WEEKLNG, WEEKLNGNUM) |>
  filter(WEEKLNG == 2 & WEEKLNGNUM != -9)

```

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##      id WEEKLNG WEEKLNGNUM
## 1 2173      2          0
## 2 3507      2          0

```

```
## 3 4197      2      0
```

```
rhfs$WEEKLNGNUM <- ifelse(rhfs$WEEKLNG == 2, 0, rhfs$WEEKLNGNUM)
```

```
# Evidence for MAR (missing at random) in MRKTVAL_R
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```
df |>
  mutate(
    mrktNA = MRKTVAL_R == -9
  ) |>
  summarize(
    survey_mean(mrktNA),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 3
##   NUMCAT_R      coef   '_se'
##   <fct>        <dbl> <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163
```

```
df |>
  filter(MRKTVAL_R != -9) |>
  summarize(
    mean_mrktval = survey_mean(MRKTVAL_R),
    median_mrktval = survey_median(MRKTVAL_R),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 5
##   NUMCAT_R      mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 1 Unit      397050.         42218.         250000         33154.
## 2 2-4 Units   569582.         33304.         425000         50491.
## 3 5-24 Units  2178309.        118540.        2000000        469164.
## 4 25-49 Units 5000928.        333704.        4000000        224403.
## 5 50+ Units  33895580.       2865340.       20000000       4987831.
```